

Cross-Manufacturer Scan Tool Implementation Reference

A deep-dive technical specification for diagnostic scan tool software engineers

Vehicle identification, CAN topology, ECU addressing, protocol negotiation,
and OEM-specific behavior required for OBD-II and enhanced diagnostics

Reference Document — Engineering Use

Based on SAE J1962, J1850, J1979, J2012, ISO 9141-2, ISO 14229, ISO 14230, ISO 15031, ISO 15765, ISO 13400, and
OEM-published service information

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1. Introduction and Scope

This document is a working reference for engineers implementing a diagnostic scan tool that must connect to, identify, and communicate with any OBD-II-equipped passenger vehicle sold in North America, Europe, or Japan since roughly 1996. It focuses on the information a scan tool needs to hold internally in order to perform three core tasks: (a) establish a working physical and data-link connection to the vehicle, (b) uniquely identify the vehicle by VIN and determine the applicable OEM, model line, model year, powertrain, and market, and (c) select the correct protocol, CAN bus, ECU addresses, and service set for generic OBD-II and enhanced manufacturer-specific diagnostics.

The document assumes the reader is familiar with the general ideas of CAN, ISO-TP segmentation, and the UDS request/response model. It does not teach those from first principles; instead it provides the lookup data, addressing schemes, decision logic, and per-manufacturer specifics that are not covered by the public standards.

A note on sources and accuracy. Much of what a cross-manufacturer scan tool needs is not standardized — it is only observable. CAN IDs for body and chassis modules, transport-protocol quirks, seed/key algorithms, and gateway behavior vary between OEMs, between model years within an OEM, and sometimes between trims. The values in this document reflect the most commonly observed and widely documented conventions at the time of writing. A production scan tool should treat every value here as a starting point that must be validated against a reference vehicle for each OEM/model-year family it intends to support.

2. What a Scan Tool Must Know

To connect to an arbitrary vehicle the scan tool must carry, or be able to derive, the following pieces of information. The rest of this document expands each of these in detail.

#	Information Required	Source / How Obtained
1	Physical pin layout of the DLC and which pins are populated	SAE J1962 connector; detect active pins on connect
2	Which OBD-II data-link protocol(s) the vehicle supports	Auto-detect via protocol scan; confirm via J1979 Mode 01 PID \$00
3	CAN bit rate and identifier length (11-bit vs 29-bit)	ISO 15765-4 initialization sequence
4	ISO-TP addressing and flow-control parameters	ISO 15765-2 defaults; per-ECU flow-control
5	VIN (17 chars) for unique identification	J1979 Mode \$09 PID \$02, or UDS \$22 \$F190
6	Manufacturer, model-family, model-year, assembly plant	Decoded from VIN (WMI / VDS / VIS)
7	Which ECUs/modules are present and their CAN IDs	Functional broadcast responses; OEM topology tables
8	Generic OBD-II service support (Modes \$01-\$0A)	J1979; Mode \$01 PID \$00/\$20/\$40/... support masks
9	Supported manufacturer-enhanced services (UDS / KWP)	Per-OEM; usually ISO 14229 or ISO 14230 semantics
10	Security access (seed/key) for protected functions	Per-OEM algorithm; \$27 service in UDS
11	DTC format (J2012) and OEM-specific code dictionary	J2012 for generic P0/B0/C0/U0; OEM for P1/B1/C1/U1

#	Information Required	Source / How Obtained
12	PID/DID dictionary and scaling formulas	J1979 Appendix B for generic; OEM databases for enhanced
13	Presence and behavior of a security gateway module	Detected by refused secured requests; OEM unlock process
14	For DoIP vehicles: Ethernet activation, routing, VIN announce	ISO 13400; detect via pin 8/12 link and UDP 13400 VIN announce

Table 2.1 — Minimum information set for a generic scan tool.

3. Physical Layer — SAE J1962 DLC

The SAE J1962 Data Link Connector (DLC) is a 16-pin trapezoidal connector, known as the Type A (12 V) or Type B (24 V) variant. Type A is used on all passenger vehicles. Type B is used on medium- and heavy-duty trucks; it has the same pin-out but a physical key that prevents a Type A plug from being inserted. The connector is required to be within 0.6 m (2 ft) of the steering column and accessible without tools.

3.1 Pin Assignments

Pin	Signal	Protocol(s)	Notes
1	OEM discretionary	Manufacturer use	Often GM SW-CAN (GMLAN single-wire, 33.3 kbps) or Ford MS-CAN+
2	J1850 Bus+ (PWM/VPW)	SAE J1850	Ford PWM uses pins 2 & 10; GM VPW uses pin 2 only
3	OEM discretionary	Manufacturer use	Ford MS-CAN (125 kbps) High on many 2003+ vehicles
4	Chassis ground	All	Connected to chassis
5	Signal ground	All	Signal reference
6	CAN High (CAN_H)	ISO 15765-4	Mandatory for HS-CAN; pair with pin 14
7	K-line	ISO 9141-2, ISO 14230	Bidirectional serial; 10.4 kbps typical
8	OEM discretionary	Manufacturer use	DoIP Activation Line on ISO 13400 vehicles; Ford MS-CAN Low on some
9	OEM discretionary	Manufacturer use	Various
10	J1850 Bus-	SAE J1850 PWM only	Ford PWM negative
11	OEM discretionary	Manufacturer use	Ford MS-CAN Low on many; Chrysler CCD/PCI on older
12	OEM discretionary	Manufacturer use	DoIP vehicles may place Ethernet TX+/RX+ here
13	OEM discretionary	Manufacturer use	DoIP Ethernet on some; OEM-specific otherwise
14	CAN Low (CAN_L)	ISO 15765-4	Mandatory for HS-CAN; pair with pin 6
15	L-line	ISO 9141-2	Optional; wake-up only on some
16	Battery +12 V (unswitched)	All	Permanent battery voltage, 4 A minimum

Table 3.1 — J1962 Type A (12 V) pin assignments.

Which pins are populated tells you which protocols to attempt. A scan tool can, and should, inspect pin presence before sending any protocol init. Voltage on pin 16 confirms the vehicle is awake and the adapter is plugged in correctly. Continuity or a terminating resistance (typically 60 Ω across CAN_H/CAN_L representing two 120 Ω terminators in parallel) on pins 6/14 indicates a HS-CAN bus is present.

3.2 Protocol-to-Pin Presence Heuristics

Pins populated	Likely protocol(s)	Typical era / OEMs
6 + 14 (CAN)	ISO 15765-4 CAN	All US vehicles 2008+; most global 2006+
6 + 14 + 8 or 12	ISO 15765-4 CAN plus DoIP	2018+ luxury; 2020+ mainstream (VW MQB Evo, BMW G-series, etc.)
2 + 10 (no CAN)	J1850 PWM	Ford / Mazda / Jaguar 1996-2003
2 only (no 10, no CAN)	J1850 VPW	GM 1996-2005; Chrysler 1996-2003
7 ($\pm$ 15), no CAN	ISO 9141-2 or ISO 14230 (KWP2000)	European / Asian 1996-2003; some EU 2003-2006
7 + 6/14 both	CAN primary, K-line fallback	Many transitional EU models 2004-2008

Table 3.2 — Inferring protocol family from DLC pin presence. Always confirm with an active protocol probe.

3.3 Electrical Characteristics

Property	HS-CAN	MS-CAN	SW-CAN (GMLAN)	J1850 VPW	J1850 PWM	K-line
Bit rate	500 kbps	125 kbps	33.3 kbps (83.3 HS)	10.4 kbps	41.6 kbps	10.4 kbps typ.
Physical	Diff. pair	Diff. pair	Single-wire	Single-wire	Diff. pair	Single-wire
Recessive V	~2.5 V both	~2.5 V both	0 V	0 V	0 V / 5 V	+12 V
Dominant V	3.5 H / 1.5 L	3.5 H / 1.5 L	4 V nom.	7.0 V	4 V / 1 V	0 V
Termination	120 Ω x2	Varies	None	None	None	Pull-up
Max nodes	~30	~30	~32	~32	~32	~2 typ.

Table 3.3 — Physical layer characteristics by protocol.

4. VIN Decoding (ISO 3779 / ISO 3780)

The Vehicle Identification Number is a 17-character string that uniquely identifies a vehicle. It is issued by the manufacturer per ISO 3779 and registered under ISO 3780. For any vehicle from model year 1981 onward, the VIN is exactly 17 characters and uses only the letters A-Z and digits 0-9, excluding I, O, and Q to avoid confusion with 1 and 0.

A scan tool should obtain the VIN from the vehicle itself, not rely on a user-entered value. The two standard ways to read it are: (a) J1979 Mode \$09 InfoType \$02, which returns VIN over OBD-II on any compliant vehicle 2006+; and (b) UDS Service \$22 Read Data By Identifier with DID \$F190, which returns VIN from any UDS-capable ECU (typically PCM, BCM, and gateway).

4.1 Structure of the 17-Character VIN

Position	Section	Meaning
1-3	WMI — World Manufacturer Identifier	Assigned by SAE (North America) and equivalent national bodies. Position 1 is the region; 2 is the manufacturer; 3 is the division or vehicle type. Small-volume manufacturers (<1000/year) encode the manufacturer in positions 12-14 instead, and use '9' in position 3.
4-8	VDS — Vehicle Descriptor Section (attributes)	Encodes model line, body style, engine, restraint system, GVWR, and other attributes. Coding is manufacturer-specific; there is no global standard for which position means what. OEM VIN-decode tables are required to interpret.
9	Check Digit	Calculated from the other 16 positions using the weighted modulus-11 algorithm (North American and Chinese VINs use this; ISO 3779 does not mandate it). The digit is 0-9 or 'X' (representing 10).
10	Model Year	Encoded as a single character (see Table 4.2). The cycle repeats every 30 years — 'A' = 1980 or 2010 or 2040. Position 7 disambiguates: it must be a digit (0-9) for model years 1980-2009 and a letter for 2010-2039.
11	Plant Code	Manufacturer-assigned identifier of the assembly plant. OEM-specific.
12-17	VIS — Vehicle Indicator Section (serial)	Production sequence number. The last 6 digits must be numeric for NHTSA-compliant North American VINs; other regions allow alphanumeric.

Table 4.1 — ISO 3779 VIN structure.

4.2 Model Year Code (Position 10)

Char	MY	Char	MY	Char	MY	Char	MY
A	1980 / 2010	L	1990 / 2020	Y	2000 / 2030	9	2009 / 2039
B	1981 / 2011	M	1991 / 2021	1	2001 / 2031	A	2040
C	1982 / 2012	N	1992 / 2022	2	2002 / 2032	—	—
D	1983 / 2013	P	1993 / 2023	3	2003 / 2033	—	—
E	1984 / 2014	R	1994 / 2024	4	2004 / 2034	—	—
F	1985 / 2015	S	1995 / 2025	5	2005 / 2035	—	—
G	1986 / 2016	T	1996 / 2026	6	2006 / 2036	—	—
H	1987 / 2017	V	1997 / 2027	7	2007 / 2037	—	—
J	1988 / 2018	W	1998 / 2028	8	2008 / 2038	—	—
K	1989 / 2019	X	1999 / 2029	—	—	—	—

Table 4.2 — Model year codes. Letters I, O, Q, U, Z and the digit 0 are never used as model year codes.

Disambiguation rule: because the model year code cycles every 30 years, a scan tool must inspect position 7 of the VIN to disambiguate. If position 7 is numeric (0-9), the vehicle is model year 1980-2009. If position 7 is alphabetic (A-Z), the vehicle is model year 2010-2039. Manufacturers adopted this convention specifically to enable this check.

4.3 WMI (World Manufacturer Identifier) — First 3 Chars

Position 1 encodes the region; positions 2-3 together identify the manufacturer and division. Small-volume manufacturers always have '9' in position 3 and append three more WMI characters in positions 12-14.

Position 1	Region
A-H	Africa
J-R (not O or Q)	Asia
S-Z	Europe
1-5	North America
6-7	Oceania
8-9	South America

Table 4.3 — VIN Position 1 region codes.

4.4 Common WMI Prefixes by OEM

WMI	Manufacturer / Division	Country
1FA / 1FB / 1FC / 1FM / 1FT / 1FD / 1ZV	Ford Motor Company	USA
2FA / 2FB / 2FC / 2FM / 2FT	Ford	Canada
3FA / 3FE	Ford	Mexico
1G1 / 1GC / 1GT / 1GK / 1GN / 1GY / 1GM / 1G6	General Motors	USA
2G1 / 2G4 / 2GC / 2GN / 2GK	General Motors	Canada
3G1 / 3GN / 3GT	General Motors	Mexico
KL1 / KL4 / KL5 / KL7 / KL8	GM Korea (Daewoo/Chevrolet)	Korea
1C3 / 1C4 / 1C6 / 2C3 / 2C4 / 3C4 / 3C6	FCA / Stellantis (Chrysler / Dodge / Jeep / Ram)	USA / Canada / Mexico
ZFA	Fiat	Italy
ZAR	Alfa Romeo	Italy
4T1 / 4T3 / 2T1 / 2T3 / 5TD / 5TF / JTD / JTE / JTJ / JTK / JTL / JTM / JTN	Toyota / Lexus	USA / Japan
JH / 1HG / 2HG / 5FN / 5J6 / 19X / 19U	Honda / Acura	Japan / USA / Canada
1N4 / 1N6 / 3N1 / 5N1 / 5N3 / JN1 / JN3 / JN6 / JN8 / JNK	Nissan / Infiniti	Japan / USA / Mexico
KM / KN / 5NM / 5NP / 5XX / 5XY	Hyundai / Kia	Korea / USA
WVW / WV1 / WV2 / WVG / 1VW / 3VW / 9BW	Volkswagen	DE / USA / Mexico / BR
WAU / WA1 / WUA / TRU	Audi	Germany / Hungary
WBA / WBS / WBX / WBY / 5UX / 5YM / 4US	BMW	Germany / USA
WDB / WDC / WDD / WDF / 4JG / 55S	Mercedes-Benz	Germany / USA
WP0 / WP1	Porsche	Germany
YV1 / YV4	Volvo Cars	Sweden / China
JF1 / JF2 / 4S3 / 4S4 / 4S6	Subaru	Japan / USA
JM1 / JM3 / 4F2 / 4F4	Mazda	Japan / USA
SAJ / SAL	Jaguar / Land Rover	UK
LSV / LFV / LGB / LJD / LVS / LFP	Chinese OEMs (Geely, Great Wall, BYD, etc.)	China
5YJ / 7SA	Tesla	USA

Table 4.4 — Selected WMI prefixes. Non-exhaustive; an OEM can own dozens of WMIs across its divisions and plants. Full authoritative list is maintained by SAE (US), Transport Canada, and equivalent national bodies.

4.5 Check Digit Algorithm (North American VINs)

A scan tool should verify VINs it receives from the vehicle. A failed check digit usually indicates a bad read or a non-NA-spec vehicle. The algorithm replaces each character with a numeric value, multiplies by a position weight, sums, and takes the result modulo 11. Result 10 is written as 'X'.

Character-to-value mapping:

```
A=1 B=2 C=3 D=4 E=5 F=6 G=7 H=8
J=1 K=2 L=3 M=4 N=5 P=7 R=9
S=2 T=3 U=4 V=5 W=6 X=7 Y=8 Z=9
0=0 1=1 2=2 3=3 4=4 5=5 6=6 7=7 8=8 9=9
(I, O, Q never appear; value for the check-digit position is not used in the sum)
```

Position weights (left to right, positions 1 to 17):

```
8, 7, 6, 5, 4, 3, 2, 10, 0, 9, 8, 7, 6, 5, 4, 3, 2
```

Sum each (value × weight), divide by 11, take the remainder. If the remainder equals the character at position 9 (treating 'X' as 10), the VIN is valid. The weight for position 9 is 0, so the check digit itself does not contribute to its own calculation.

Worked example — VIN 1M8GDM9AXKP042788 (the SAE reference VIN):

```
Chars: 1 M 8 G D M 9 A X K P 0 4 2 7 8 8
Values: 1 4 8 7 4 4 9 1 - 2 7 0 4 2 7 8 8
Weights: 8 7 6 5 4 3 2 10 0 9 8 7 6 5 4 3 2
Products: 8 28 48 35 16 12 18 10 - 18 56 0 24 10 28 24 16
Sum = 351. 351 mod 11 = 10 → 'X'. Position 9 is 'X'. Valid.
```

4.6 Pseudocode — Compact VIN Decoder

```
def decode_vin(vin):
    assert len(vin) == 17 and all(c in VIN_CHARS for c in vin)
    wmi = vin[0:3] # manufacturer lookup
    vds = vin[3:8] # attributes (OEM-specific)
    chk = vin[8] # check digit
    myc = vin[9] # model-year char
    plant = vin[10] # plant code
    serial = vin[11:17]
    region = REGION_MAP[vin[0]]
    oem = WMI_MAP.get(wmi) or WMI_MAP.get(vin[0:2])
    year = resolve_year(myc, vin[6]) # pos-7 disambiguation
    ok = validate_check_digit(vin)
    model = oem.decode_vds(vds, year) # delegate to OEM table
    return VehicleId(oem=oem, model=model, year=year,
                    plant=plant, serial=serial, check_ok=ok)
```

5. OBD-II Protocol Family Overview

OBD-II, mandated in the US from 1996 model year and (as EOBD/JOBD) in most of the world thereafter, is a family of five data-link protocols wrapped in a common application layer defined by SAE J1979 / ISO 15031-5. A scan tool must implement all five to reach the full 1996-2024 vehicle population; a tool that only targets 2008+ US-market vehicles can get away with just CAN.

Protocol	Standard	Bit rate	Physical	Typical use
ISO 15765-4 CAN (11-bit)	ISO 11898 / 15765	500 kbps	Diff. pair	All US 2008+, most global 2006+
ISO 15765-4 CAN (29-bit)	ISO 11898 / 15765	500 kbps	Diff. pair	Heavy duty; some EU and GM
ISO 9141-2	ISO 9141-2:1994	10.4 kbps	Single-wire (K)	EU / Asian 1996-2004
ISO 14230-4 (KWP2000)	ISO 14230	10.4 kbps (fast-init) or slow-init	Single-wire (K)	EU / Asian 2000-2008
SAE J1850 VPW	SAE J1850	10.4 kbps	Single-wire	GM 1996-2005; Chrysler 1996-2003
SAE J1850 PWM	SAE J1850	41.6 kbps	Differential pair	Ford 1996-2003 (and Mazda / Jaguar)
DoIP (ISO 13400)	ISO 13400 + UDS	100 Mbps / 1 Gbps	Ethernet over pin 8/12 or 3/11	2018+ luxury; 2020+ mainstream

Table 5.1 — OBD-II protocol family.

5.1 Application Layer — J1979 / ISO 15031-5

On top of any of the five data-link protocols, the generic OBD-II application layer (J1979) defines request/response messages organized into numbered services called *modes*. Every OBD-II vehicle must support a minimum subset of these modes for emissions-related ECUs. The modes are summarized in Chapter 9; the essential fact here is that the same 10 modes exist on top of J1850, K-line, and CAN, with only minor framing differences. A scan tool's generic OBD-II layer can be written once and adapted per-transport.

5.2 Enhanced Diagnostics Application Layer — UDS / KWP

Beyond generic OBD-II, every manufacturer exposes *enhanced* diagnostics to its factory scan tools. These use either ISO 14230 (KWP2000) on pre-2007 vehicles or ISO 14229 (UDS) on 2007+ vehicles. UDS is transport-independent: it runs over CAN (via ISO-TP), over DoIP, over FlexRay, and in principle over LIN or K-line. A cross-manufacturer tool therefore needs a single UDS client that can be pointed at any of those transports.

5.3 Quick Comparison of Common Modern Stacks

Vehicle class (approx.)	DLC	Transport	App layer (generic)	App layer (enhanced)
US 1996-2000 domestic	J1962	J1850 VPW/PWM	J1979	SAE J2190 / OEM
EU / JP 1996-2003	J1962	ISO 9141-2	J1979	KWP2000 (ISO 14230)
Global 2004-2007 transition	J1962	ISO 14230 / CAN	J1979	KWP2000 or UDS
Global 2008-2017	J1962	CAN 500 kbps	J1979 over ISO-TP	UDS (ISO 14229)
2018+ luxury, 2020+ mass-market	J1962 + DoIP	CAN + Ethernet	J1979 over ISO-TP / DoIP	UDS over CAN and DoIP

Table 5.2 — Protocol stacks by vehicle era.

6. Protocol Auto-Detection Logic

The most robust way to identify a vehicle's data-link protocol is to attempt each one in order of decreasing likelihood and stop on the first valid response. The recommended probe is a J1979 Mode \$01 PID \$00 request: every OBD-II-compliant vehicle must respond to this with a bitmask of supported PIDs. A valid response on a given transport confirms both the transport and the presence of an emissions ECU. The order below is a reasonable default; adapters like the ELM327 implement essentially this sequence.

6.1 Recommended Probe Sequence

Step	Protocol to try	What you send	What a valid response looks like
1	ISO 15765-4 CAN 11-bit, 500 kbps	Functional \$7DF → 02 01 00	Any CAN ID in \$7E8-\$7EF answering 06 41 00 BB BB BB BB
2	ISO 15765-4 CAN 29-bit, 500 kbps	Functional \$18DB33F1 → 02 01 00	Response ID \$18DAF1xx with 06 41 00 BB BB BB BB
3	ISO 15765-4 CAN 11-bit, 250 kbps	Same as step 1, lower rate	Medium/heavy duty, some EV charging ECUs
4	ISO 14230-4 fast init (KWP2000)	Fast-init sequence + 01 00 (header \$C2 \$33 \$F1)	Header from ECU + 41 00 BB BB BB BB
5	ISO 14230-4 5-baud init	5-baud address \$33, synchronize at 0x55, get key bytes	Key bytes \$8F \$E9 typical; then 41 00 ...
6	ISO 9141-2 5-baud init	Same 5-baud init, different key bytes	Key bytes \$08 \$08 or \$94 \$94 typical
7	J1850 VPW (GM/Chrysler)	Header \$68 \$6A \$F1, PID 01 00	Header \$48 \$6B \$10 or similar + 41 00 ...
8	J1850 PWM (Ford)	Header \$61 \$6A \$F1, PID 01 00	Header \$41 \$6B \$10 + 41 00 ...

Table 6.1 — Protocol probe sequence. Each step should time out within ~500 ms before advancing.

6.2 Decision Tree

1. Read DLC pin presence via adapter's voltage probe.
 - If pins 6 & 14 show CAN-like impedance (~60 Ω between them): start CAN probe.
 - Else if pin 7 has ~+12 V pull-up: likely K-line (ISO 9141 / 14230).
 - Else if pins 2 & 10 present: J1850 PWM.
 - Else if pin 2 only: J1850 VPW.
2. On CAN: send 11-bit functional \$7DF → 02 01 00 00 00 00 00 00.
Wait 200 ms for any response in \$7E8-\$7EF. If received, 11-bit CAN at 500 kbps confirmed.
3. If no response, send 29-bit functional \$18DB33F1 → 02 01 00 ...
If received, 29-bit CAN at 500 kbps confirmed.
4. If no CAN response at 500 kbps, retry at 250 kbps (rare; some heavy duty).
5. If K-line: try KWP2000 fast-init first (faster, supported by more 2003+ EU/JP). On failure, fall back to 5-baud init.
6. Parse Mode \$01 PID \$00 response to learn supported-PID bitmask.
7. Issue Mode \$09 PID \$02 to retrieve VIN. Decode to determine OEM and model year.
8. Using OEM and model year, look up enhanced protocol (UDS vs KWP), expected CAN topology, and module address table.
9. Optionally activate DoIP: if pins 8/12 show Ethernet link, send UDP broadcast to 13400 and listen for VIN announce.

6.3 Confirming the Protocol — Mode \$01 PID \$00 Bitmask

PID \$00 returns a 4-byte bitmask where bit 31 (MSB of byte 0) corresponds to PID \$01, bit 30 to PID \$02, and so on down to bit 0 = PID \$20. If bit 0 is set, PID \$20 is supported, and PID \$20 itself returns the bitmask for PIDs \$21-\$40. This chaining continues: PID \$40 returns \$41-\$60, PID \$60 returns \$61-\$80, PID \$80 returns \$81-\$A0, PID \$A0 returns \$A1-\$C0, PID \$C0 returns \$C1-\$E0. A scan tool should query the full chain to build the complete supported-PID set before trusting any specific PID read.

7. ISO 15765-4 CAN — The Modern Default

ISO 15765-4 constrains the generic CAN standards (ISO 11898-1 for data link, ISO 11898-2 for physical) to the four variants actually used for OBD-II. It also fixes addressing conventions and timing parameters. For any US vehicle 2008 or newer, and most non-US vehicles 2006 or newer, this is the protocol the scan tool will speak.

7.1 The Four Variants

Variant	Identifier length	Bit rate	Typical use
11-bit @ 500 kbps	11-bit standard frame	500 kbps	Default for nearly all light-duty passenger vehicles
11-bit @ 250 kbps	11-bit standard frame	250 kbps	Some commercial / medium-duty
29-bit @ 500 kbps	29-bit extended frame	500 kbps	Heavy-duty passenger (some Ford Super Duty, some GM HD); Harmonized with SAE J1939 addressing
29-bit @ 250 kbps	29-bit extended frame	250 kbps	Heavy-duty / commercial

Table 7.1 — ISO 15765-4 CAN variants.

7.2 11-Bit OBD-II Addressing

The 11-bit scheme uses a single fixed functional address and a contiguous range of physical addresses, one pair per ECU. The same pattern is used by virtually every 11-bit CAN implementation worldwide.

Address	Direction	Role
\$7DF	Tester → vehicle (broadcast)	Functional request; all emissions-related ECUs listen and may answer
\$7E0	Tester → ECU	Physical request to ECU #1 — by convention the Engine/PCM
\$7E1	Tester → ECU	Physical request to ECU #2 — typically Transmission (TCM)
\$7E2 through \$7E7	Tester → ECU	Physical request to ECUs #3 through #8
\$7E8	ECU → tester	Response from ECU #1 (PCM). Paired with request \$7E0.
\$7E9	ECU → tester	Response from ECU #2 (TCM). Paired with \$7E1.
\$7EA through \$7EF	ECU → tester	Responses from ECUs #3 through #8

Table 7.2 — 11-bit OBD-II CAN addressing. Request ID + 8 = response ID. Only ECUs with generic OBD-II responsibilities appear in this range; body, chassis, and infotainment modules use OEM-specific ranges outside \$7Ex.

7.3 29-Bit OBD-II Addressing

The 29-bit scheme reserves three fields within the extended identifier: a priority / protocol field (upper 8 bits), a target address (next 8 bits), and a source address (lower 8 bits). The tester's source address is always \$F1. The special target address \$33 means 'all emissions ECUs' (functional broadcast).

29-bit CAN ID	Meaning
\$18 DB 33 F1	Functional request from tester (\$F1) to all emissions ECUs (\$33)
\$18 DA xx F1	Physical request from tester to specific ECU, where xx = ECU address
\$18 DA F1 xx	Physical response from ECU to tester

Table 7.3 — 29-bit OBD-II CAN addressing. Typical ECU target bytes: \$00/\$01 Engine, \$02 ABS, \$03 TCM, \$07 BCM, \$60+ infotainment. OEM-specific.

7.4 Frame Format Recap

Field	11-bit frame	29-bit frame
SOF	1 bit dominant	1 bit dominant
Identifier	11 bits	11 bits (base) + 18 bits (ext)
RTR/SRR	1 bit	SRR + IDE + RTR
DLC	4 bits (0-8 for classic CAN)	4 bits
Data	0-8 bytes	0-8 bytes
CRC	15 bits + delim	15 bits + delim
ACK	1 bit + delim	1 bit + delim
EOF	7 bits recessive	7 bits recessive

Table 7.4 — CAN frame fields (classic CAN). CAN-FD extends DLC to 15 (64 bytes) and adds a separate data-phase bit rate, but is not yet standardized for OBD-II.

8. ISO-TP (ISO 15765-2) Transport

A classic CAN frame carries at most 8 bytes of payload. Diagnostic messages are often longer — a VIN response alone is 20 bytes. ISO 15765-2, called ISO-TP or the Transport Protocol, segments messages longer than 7 bytes across multiple frames. Every byte a scan tool sends or receives on CAN goes through ISO-TP.

8.1 Frame Types and the PCI Byte

Each CAN frame's first data byte is the Protocol Control Information (PCI). Its upper nibble identifies the frame type:

Type	Upper nibble	Name	Lower nibble / following bytes
0	\$0	Single Frame (SF)	Low nibble = length (0-7). Bytes 1-7 = payload.
1	\$1	First Frame (FF)	Low nibble + next byte = 12-bit total length. Bytes 2-7 = first 6 payload bytes.
2	\$2	Consecutive Frame (CF)	Low nibble = sequence number 0-F, wrapping. Bytes 1-7 = 7 more payload bytes.
3	\$3	Flow Control (FC)	Low nibble = flow status. Bytes 1-2 = block size and STmin.

Table 8.1 — ISO-TP frame types and PCI byte layout.

8.2 Example — Reading the VIN (~20 bytes) over ISO-TP

Tester → PCM (\$7E0) — request VIN, single frame (3 bytes payload):

```
7E0 02 09 02 00 00 00 00 00 ; SF len=2, service 09, PID 02
```

PCM → Tester (\$7E8) — first frame announcing 20-byte response:

```
7E8 10 14 49 02 01 31 46 41 ; FF len=0x0014 (20), then 6 payload bytes
      ; "49 02 01" = positive resp to 09 02 + data item count 1
      ; "31 46 41" = '1', 'F', 'A' (first 3 VIN chars)
```

Tester → PCM (\$7E0) — flow control, continue to send:

```
7E0 30 00 00 00 00 00 00 00 ; FC, flag 0 (CTS), BS 0 (all), STmin 0
```

PCM → Tester (\$7E8) — consecutive frames carrying remainder of VIN:

```
7E8 21 50 32 46 41 33 30 41 ; CF seq 1: 'P', '2', 'F', 'A', '3', '0', 'A'
7E8 22 36 30 32 31 30 31 32 ; CF seq 2: '6', '0', '2', '1', '0', '1', '2'
```

Total useful payload extracted: **49 02 01 || '1FAP2FA30A6021012'** — that is, service \$49, positive response to \$09 \$02, one data item, followed by the 17-character VIN. The ISO-TP layer assembled the three CAN frames into a 20-byte diagnostic payload; the J1979 layer stripped the service/subservice/count bytes; the VIN-decode layer consumed the remaining ASCII.

8.3 Flow Control Parameters (ECU → Tester)

Byte	Name	Meaning
0 (PCI hi)	\$3 (FC)	Fixed — Flow Control frame type
0 (PCI lo)	FS — Flow Status	0 = Continue To Send (CTS), 1 = Wait, 2 = Overflow/abort
1	BS — Block Size	How many CFs the sender may transmit before waiting for the next FC. 0 = no limit.
2	STmin — Separation Time	Minimum delay between CFs. \$00-\$7F = N milliseconds; \$F1-\$F9 = 100-900 μs.

Table 8.2 — Flow Control frame fields. Scan tools should honor the ECU's STmin strictly; many ECUs will drop the session if CFs arrive too fast.

8.4 ISO-TP Timing Parameters

Timer	Applies to	Value	What it measures
N_As	Sender	1000 ms max	Time to transmit a single CAN frame
N_Ar	Receiver	1000 ms max	Time to transmit its FC frame
N_Bs	Sender	1000 ms max	Waiting for FC after FF or BS-th CF
N_Br	Receiver	N_Bs-N_Ar max	Receiver delay before sending FC
N_Cs	Sender	N_Cr-N_As max	Sender delay between CFs (subject to STmin)
N_Cr	Receiver	1000 ms max	Waiting for next CF

Table 8.3 — ISO 15765-2 network timing parameters.

9. OBD-II Diagnostic Services (SAE J1979)

SAE J1979, aligned with ISO 15031-5, defines ten *service modes* for generic emissions-related diagnostics. Each request byte is one of these mode numbers; the ECU responds with the mode + \$40 on success, or with \$7F + mode + NRC (negative response code) on failure. A scan tool's generic OBD-II layer needs to support all ten, even if only PCM Mode \$01 is commonly used by consumers.

Mode	Name	Purpose	Positive response
\$01	Show current data	Read live sensor data by PID (see Chapter 13)	\$41 + PID + data
\$02	Show freeze frame data	Read sensor snapshot captured when a DTC was stored	\$42 + PID + frame# + data
\$03	Show stored DTCs	Retrieve confirmed fault codes currently illuminating the MIL	\$43 + count + DTC list
\$04	Clear DTCs and emission data	Erase all stored DTCs, freeze frame, readiness, O2 learned values	\$44 (no payload)
\$05	O2 sensor monitoring results	Legacy non-CAN only. Replaced by Mode \$06 on CAN vehicles.	\$45 + test ID + data
\$06	On-board monitoring test results	Continuous / non-continuous monitor results (the 'real' way to see catalyst efficiency, misfire counts, EVAP leak, etc.)	\$46 + MID + TID + data
\$07	Show pending DTCs	Failures detected during the current or last drive cycle but not yet confirmed by a second failure — useful for drive-cycle verification	\$47 + DTC list
\$08	Request control of on-board system	Bidirectional control — e.g., EVAP leak test command	\$48 + TID + data
\$09	Request vehicle information	VIN (PID \$02), Calibration IDs (PID \$04), CVN (PID \$06), ECU name (PID \$0A), performance tracking (PID \$08)	\$49 + PID + data
\$0A	Permanent DTCs	DTCs that cannot be cleared by Mode \$04; require a successful drive cycle with no fault to self-clear. 2010+ US-market.	\$4A + DTC list

Table 9.1 — J1979 service modes.

9.1 Negative Response Codes (NRC)

When an ECU cannot or will not fulfill a request, it returns \$7F followed by the original service byte and an NRC byte. Common J1979 NRCs:

NRC	Meaning
\$10	General reject
\$11	Service not supported
\$12	Sub-function / PID not supported
\$13	Incorrect message length or invalid format
\$21	Busy — repeat request
\$22	Conditions not correct (e.g., engine running required)
\$31	Request out of range
\$78	Response pending — ECU is working, wait

Table 9.2 — Common J1979 / UDS negative response codes.

10. UDS — ISO 14229 Services Reference

Unified Diagnostic Services, ISO 14229, is the application layer used by every modern OEM for enhanced diagnostics, module programming, and configuration. Its service set is a strict superset of J1979's in function, though the byte-level encoding is different. A cross-manufacturer scan tool needs a UDS client that can target any ECU on any transport.

Service IDs below \$40 are requests; the positive response always uses SID + \$40. A negative response is \$7F + SID + NRC, same convention as J1979. The table below lists the services most relevant to scan tool work.

SID	Service Name	Purpose / Scan-tool Use
\$10	Diagnostic Session Control	Switch between default, programming, extended-diag, and safety-system sessions. Most non-default services require extended session (\$10 03).
\$11	ECU Reset	Hard reset (01), key-off-on (02), soft (03), enable-rapid-power-shutdown (04), disable-rapid-power-shutdown (05).
\$14	Clear Diagnostic Information	UDS equivalent of Mode \$04, but scoped: can clear a specific DTC group. Group \$FFFFFF clears all.
\$19	Read DTC Information	20+ sub-functions. Most used: \$02 reportDTCByStatusMask, \$04 reportDTCSnapshotRecordByDTCNumber, \$06 reportDTCExtendedDataRecordByDTCNumber, \$0A reportSupportedDTC.
\$22	Read Data By Identifier	The workhorse. Read a 16-bit DID. VIN = \$F190, SW version = \$F195, ECU hw serial = \$F18C, calibration ID = \$F191, manufacturer spare-part number = \$F187. OEMs publish their DID dictionaries.
\$23	Read Memory By Address	Raw memory read. Arguments include an AddressAndLengthFormat byte, an address, and a length. Used for module analysis.
\$24	Read Scaling Data By Identifier	Returns formatting / scaling metadata for a given DID.
\$27	Security Access	Two-step seed/key. Sub-function N (odd) = request seed; N+1 = send key. Algorithm is OEM/level-specific — see Chapter 15.
\$28	Communication Control	Enable/disable tx or rx of normal CAN traffic during programming. Sub-function 03 = disable rx & tx; 00 = enable both.
\$29	Authentication (newer UDS 2020)	Certificate-based replacement for \$27. Only on very recent vehicles.
\$2A	Read Data By Periodic Identifier	Stream DIDs at slow, medium, or fast periodicity.
\$2C	Dynamically Define Data Identifier	Assemble a custom DID from pieces of other DIDs to allow single-request reads.
\$2E	Write Data By Identifier	Update a writable DID. Needed for VIN write after ECU swap, odometer write, PATS / immobilizer coding.
\$2F	Input Output Control By Identifier	Bidirectional control. Force actuators: fuel pump, EVAP purge, injector balance, cooling fan, transmission solenoid.
\$31	Routine Control	Start (\$01), stop (\$02), request result (\$03) of a named routine. Covers: compression test, relearn procedures, calibration resets, ABS bleed cycles, throttle-body idle relearn, VIN write routines.
\$34	Request Download	Initiate a tester-to-ECU download. Used during reflashing.

SID	Service Name	Purpose / Scan-tool Use
\$35	Request Upload	Initiate an ECU-to-tester upload. Used for memory dump / backup.
\$36	Transfer Data	Send/receive data blocks during a \$34/\$35 transfer.
\$37	Request Transfer Exit	Close the transfer and verify checksum.
\$38	Request File Transfer	File-level transfer, newer than \$34/\$36/\$37. Rare.
\$3D	Write Memory By Address	Raw memory write. Requires security access.
\$3E	Tester Present	Keep-alive. Must be sent every ~2 s to maintain a non-default session. Sub-function \$80 = suppress response for efficiency.
\$83	Access Timing Parameter	Negotiate custom session timing.
\$84	Secured Data Transmission	Encrypted payload (used with security gateways).
\$85	Control DTC Setting	Pause DTC logging during noisy procedures. \$01 = on, \$02 = off.
\$86	Response On Event	ECU sends asynchronous notifications when an event occurs.
\$87	Link Control	Switch CAN to a non-standard bit rate for programming (e.g., 1 Mbps).

Table 10.1 — ISO 14229 (UDS) service identifiers. A full scan tool implements \$10, \$11, \$14, \$19, \$22, \$27, \$2E, \$2F, \$31, \$3E, \$85 at minimum; flashing adds \$34-\$37 and \$28.

10.1 Key UDS Data Identifiers (DIDs)

DID	Name	Meaning
\$F180	BootSoftwareIdentificationDataIdentifier	Bootloader ID
\$F181	ApplicationSoftwareIdentificationDataIdentifier	Application SW ID
\$F182	ApplicationDataIdentificationDataIdentifier	Application data / cal
\$F183	BootSoftwareFingerprintDataIdentifier	Boot fingerprint
\$F184	ApplicationSoftwareFingerprintDataIdentifier	App SW fingerprint
\$F186	ActiveDiagnosticSessionDataIdentifier	Read active session (UDS \$10)
\$F187	VehicleManufacturerSparePartNumber	OEM spare part number
\$F188	VehicleManufacturerECUSoftwareNumberDataIdentifier	ECU SW number
\$F189	VehicleManufacturerECUSoftwareVersionNumberDataIdentifier	ECU SW version
\$F18A	SystemSupplierIdentifierDataIdentifier	Supplier (e.g., Bosch, Conti)
\$F18B	ECUManufacturingDateDataIdentifier	Manufacturing date YYMMDD
\$F18C	ECUSerialNumberDataIdentifier	ECU hardware serial
\$F190	VIN Data Identifier	17-character VIN
\$F191	VehicleManufacturerECUHardwareNumberDataIdentifier	Hardware part number
\$F192	SystemSupplierECUHardwareNumberDataIdentifier	Supplier HW number
\$F193	SystemSupplierECUHardwareVersionNumberDataIdentifier	Supplier HW version
\$F194	SystemSupplierECUSoftwareNumberDataIdentifier	Supplier SW number
\$F195	SystemSupplierECUSoftwareVersionNumberDataIdentifier	Supplier SW version
\$F197	SystemNameOrEngineTypeDataIdentifier	Human-readable name
\$F198	RepairShopCodeOrTesterSerialNumberDataIdentifier	Last writer
\$F199	ProgrammingDateDataIdentifier	Last flash date
\$F19D	ECUInstallationDateDataIdentifier	Installed in vehicle on
\$F19E	ODXFileDataIdentifier	Associated ODX file name

Table 10.2 — Standard UDS DIDs in the \$F1xx range (ISO 14229-1 Annex F).

11. ECU/Module Naming and Typical Addresses

The number, names, and CAN addresses of modules in a vehicle vary by OEM, model, year, and even trim. The table below lists the common function-to-module mapping that holds across most modern vehicles. Chapter 14 provides OEM-specific address tables.

Abbreviation	Full Name	Function
PCM / ECM	Powertrain / Engine Control Module	Engine management. Generic OBD-II primary responder. Always at \$7E0/\$7E8 on 11-bit CAN.
TCM	Transmission Control Module	Automatic transmission control. Often integrated into PCM on smaller vehicles. \$7E1/\$7E9 when separate.

Abbreviation	Full Name	Function
ABS / EBCM	Antilock Brake / Electronic Brake Control	Wheel speed sensors, ABS solenoids, stability control. Often extended with ESC / VSC functions.
SRS / ACM / RCM	Supplemental Restraints / Airbag Control	Airbag firing, seatbelt pretensioners, rollover detection. Crash-fused; usually read-only via OBD.
BCM / BCMII	Body Control Module	Exterior lighting, wipers, locks, power windows (some), interior lighting, horn.
IPC / IC	Instrument Panel Cluster / Instrument Cluster	Speedo / tach / gauges, driver info display, warning chime, odometer storage.
HVAC / ATC	Heating, Ventilation, A/C Control	Automatic climate control, blower, mode doors, rear defrost.
EPS / PSCM	Electric Power Steering / Power Steering Control	Torque assist motor, steering angle sensor interface.
TPMS / RCDLR	Tire Pressure Monitor	Often integrated in BCM on modern vehicles. Older Ford has dedicated RCDLR.
Radio / APIM / IHU	Infotainment Head Unit	Audio, nav, Bluetooth, Apple CarPlay / Android Auto. Ford SYNC = APIM.
TCU / OCU / TCB	Telematics Control Unit	Cellular modem (OnStar, BlueLink, UVO, ConnectedDrive, FordPass, Mercedes me).
GWM / CGM / CGW	Central Gateway / Secure Gateway	Router between CAN buses; firewall for diagnostic requests; authentication authority.
DIM / HUD	Driver Information / Head-Up Display	Driver-side information displays separate from IPC.
PCI / RFA / KPM	Keyless / PATS / Immobilizer	Key authentication, remote-start authorization. PATS on Ford, PASSLOCK on GM, SKIM on Chrysler.
PTC / FTC / FCC	Powertrain Isolation / Hybrid / EV	Battery management, inverter control, DC-DC on hybrids/EVs.
ACC / FCWS / LKAS	Driver Assist modules	Adaptive cruise, forward-collision warning, lane-keep. Often combined as 'ADAS'.
SAS / SWM	Steering Angle Sensor / Steering Wheel Module	Steering position (needed by ABS and ACC) plus column controls.
RFH / SDM	Rear Function / Sensing Diagnostic	Blind-spot, rear park assist, trunk module.

Table 11.1 — Common ECU functional naming. Different OEMs use different three-letter acronyms for the same physical box — e.g., BCM (GM) ≡ BEM (Chrysler) ≡ GEM/SJB (Ford) ≡ BCU / ZGM (Mercedes).

11.1 How ECU Presence Is Discovered

There are three practical ways to discover which modules a given vehicle has:

- 1. Functional broadcast probing.** Send \$7DF → 02 10 03 (UDS extended session) or simply \$7DF → 01 00 (J1979) and collect every responder. Each unique response ID in \$7E8-\$7EF is a module that speaks OBD-II. For non-OBD modules on the same bus, probe outside the \$7Ex range using OEM-known IDs.
- 2. Gateway inventory.** On vehicles with a Central Gateway that implements ISO 13400 (DoIP) routing, send a Vehicle Identification Request to get back a list of logical address / routing table entries — the gateway tells the tool which modules exist behind which CAN bus.
- 3. OEM topology tables.** For each OEM/model/year, maintain a hand-curated table of expected modules and their CAN IDs. The tool probes each expected ID; present / absent is recorded. This is the approach most professional scan tools use in practice because it is faster and more reliable than open discovery.

11.2 Per-Module Addressing Beyond OBD-II Range

Modules outside the generic OBD-II address range (\$7E0-\$7EF on 11-bit CAN) use OEM-specific CAN IDs. These are not standardized and must be learned per OEM. The general pattern is:

Convention	Example OEMs using it
Request ID = Response ID – \$08 (same offset as OBD-II)	GM, Hyundai/Kia, FCA on most bodies
Response ID = Request ID + \$400	Some BMW and Volvo
Request ID and Response ID unrelated (looked up per-module)	Ford (many modules), VW/Audi (legacy TP 2.0), Mercedes
29-bit physical pair with target/source swapped	Any OEM using 29-bit for non-OBD modules

Table 11.2 — Common ID-pair conventions outside the generic OBD-II range.

12. DTC Structure and Decoding (SAE J2012)

A Diagnostic Trouble Code (DTC) is a 2- or 3-byte identifier an ECU logs when it detects a fault. SAE J2012 defines the textual format used on scan tool displays and the meaning of the generic code space. On CAN, the 16-bit (J1979) and 24-bit (UDS) forms are both used.

12.1 Textual Form — The ABXXX Layout

Every DTC is displayed as a letter followed by four hex digits. The leading letter identifies the subsystem; the first digit after the letter identifies the code range as SAE-generic or manufacturer-specific; the remaining three digits are the fault number.

Letter	Subsystem
P	Powertrain — engine, transmission, emissions
B	Body — airbags, power windows, lighting, HVAC
C	Chassis — ABS, steering, suspension
U	Network / Communication — lost message from another ECU, bus-off

Table 12.1 — DTC first-letter meanings.

First digit	Meaning
0	SAE / ISO generic — meaning defined by J2012; same across all OEMs
1	Manufacturer-specific — meaning per OEM table; often overlaps generic meaning
2	Generic (secondary SAE range, used when 0xxx is exhausted — e.g., P2xxx)
3	Mixed — P3xxx generic; B3/C3/U3 manufacturer. Modern standards place more in generic.

Table 12.2 — DTC second-character (generic vs OEM) meaning.

12.2 Binary Encoding

The 16-bit binary form packs the two-character prefix into the top four bits:

Top 2 bits	Letter	Next 2 bits	First digit
00	P (Powertrain)	00	0 — generic
01	C (Chassis)	01	1 — manufacturer
10	B (Body)	10	2 — generic
11	U (Network)	11	3 — mixed

Table 12.3 — 16-bit DTC encoding.

Example: the bytes **01 33** decode as binary 0000 0001 0011 0011. Top 2 bits = 00 → P. Next 2 bits = 00 → generic. Remaining 12 bits = 0x133 → '133'. Full DTC: **P0133** — O2 Sensor Circuit Slow Response (Bank 1 Sensor 1).

12.3 UDS 24-bit DTC Form

UDS (ISO 14229) extends DTCs to 24 bits by appending a failure-type byte: **DTCHighByte - DTCMidByte - FTB**. The first two bytes carry the J2012 code; the third byte describes the symptom (fault type). Common FTB values:

FTB	Symptom
\$00	No sub-type information
\$11	Circuit short to ground
\$12	Circuit short to battery
\$13	Circuit open
\$14	Circuit short
\$16	Below threshold
\$17	Above threshold
\$1C	Voltage out of range
\$23	Signal stuck low
\$24	Signal stuck high
\$29	Signal invalid
\$2F	Signal drift / bias
\$49	Internal ECU fault
\$54	Missing calibration
\$55	Not programmed
\$62	Signal compare mismatch
\$64	Signal plausibility
\$71	Actuator stuck
\$81	Message count incorrect
\$87	Missing message
\$88	Bus off
\$92	Performance / incorrect operation
\$96	Component internal failure
\$98	Component or system over-temperature

Table 12.4 — Common UDS failure-type bytes (non-exhaustive).

12.4 DTC Status Byte

UDS returns each DTC with an 8-bit status byte. Each bit has a specific meaning defined in ISO 14229. A scan tool presenting DTCs to the user should decode and display the relevant bits; otherwise a 'pending' code and a 'confirmed' code look identical on screen.

Bit	Name	Meaning when set
0	testFailed	Most recent test result: failed

Bit	Name	Meaning when set
1	testFailedThisOperationCycle	Failed at least once this drive cycle
2	pendingDTC	Failed this or previous cycle, not yet confirmed
3	confirmedDTC	Confirmed DTC (MIL candidate)
4	testNotCompletedSinceLastClear	Monitor hasn't run since last Mode \$04
5	testFailedSinceLastClear	Failed at least once since last clear
6	testNotCompletedThisOperationCycle	Monitor hasn't run this cycle
7	warningIndicatorRequested	MIL / warning lamp is being requested

Table 12.5 — UDS DTC status byte bits (ISO 14229-1).

13. Mode 01 PID Reference and Scaling

Mode \$01 PIDs are the standardized live-data sensors. A scan tool queries PID \$00 first to learn the supported-PID bitmask, then requests only supported PIDs. The formulas below convert raw response bytes into engineering units. A, B, C, D refer to the first, second, third, and fourth data bytes of the response. This is an abbreviated list — J1979 Appendix B defines all ~200 PIDs.

PID	Description	Bytes	Formula	Units
\$00	Supported PIDs \$01-\$20	4	Bitmask	-
\$01	Monitor status since DTCs cleared, MIL	4	A.7 = MIL on/off; A.0-6 = DTC count; B-D = monitor readiness	-
\$02	Freeze frame DTC that triggered storage	2	See Mode \$02	-
\$03	Fuel system status	2	Bitmask per bank	-
\$04	Calculated engine load	1	$A * 100 / 255$	%
\$05	Engine coolant temperature	1	$A - 40$	°C
\$06	Short-term fuel trim, bank 1	1	$(A - 128) * 100 / 128$	%
\$07	Long-term fuel trim, bank 1	1	$(A - 128) * 100 / 128$	%
\$08	Short-term fuel trim, bank 2	1	$(A - 128) * 100 / 128$	%
\$09	Long-term fuel trim, bank 2	1	$(A - 128) * 100 / 128$	%
\$0A	Fuel rail pressure (gauge)	1	$A * 3$	kPa
\$0B	Intake manifold absolute pressure	1	A	kPa
\$0C	Engine RPM	2	$(256 * A + B) / 4$	rpm
\$0D	Vehicle speed	1	A	km/h
\$0E	Timing advance (cyl 1)	1	$(A - 128) / 2$	° before TDC
\$0F	Intake air temperature	1	$A - 40$	°C
\$10	MAF air flow rate	2	$(256 * A + B) / 100$	g/s
\$11	Throttle position	1	$A * 100 / 255$	%

PID	Description	Bytes	Formula	Units
\$13	O2 sensors present (2-bank)	1	Bitmask	-
\$14-\$1B	O2 sensor 1-8 voltage / trim	2	$A/200 \text{ V}; (B-128)*100/128 \%$	V / %
\$1C	OBD standard compliance	1	Enum — 01=CARB, 02=EPA, 03=OBD&OBD-II, ...	-
\$1F	Run time since engine start	2	$256*A + B$	s
\$20	Supported PIDs \$21-\$40	4	Bitmask	-
\$21	Distance traveled with MIL on	2	$256*A + B$	km
\$22	Fuel rail pressure (relative to manifold)	2	$(256*A + B) * 0.079$	kPa
\$23	Fuel rail gauge pressure (diesel, direct inject)	2	$(256*A + B) * 10$	kPa
\$2C	Commanded EGR	1	$A * 100 / 255$	%
\$2D	EGR error	1	$(A - 128) * 100 / 128$	%
\$2E	Commanded evaporative purge	1	$A * 100 / 255$	%
\$2F	Fuel tank level	1	$A * 100 / 255$	%
\$30	Warm-ups since DTCs cleared	1	A	count
\$31	Distance traveled since DTCs cleared	2	$256*A + B$	km
\$33	Barometric pressure	1	A	kPa
\$40	Supported PIDs \$41-\$60	4	Bitmask	-
\$42	Control module voltage	2	$(256*A + B) / 1000$	V
\$43	Absolute load value	2	$(256*A + B) * 100 / 255$	%
\$44	Commanded equivalence ratio (λ)	2	$(256*A + B) / 32768$	ratio
\$45	Relative throttle position	1	$A * 100 / 255$	%
\$46	Ambient air temperature	1	$A - 40$	°C
\$4D	Time with MIL on	2	$256*A + B$	min
\$4E	Time since DTCs cleared	2	$256*A + B$	min
\$51	Fuel type	1	Enum — 01=gasoline, 04=diesel, 08=electric, ...	-
\$5B	Hybrid/EV battery pack remaining life	1	$A * 100 / 255$	%
\$5C	Engine oil temperature	1	$A - 40$	°C
\$60	Supported PIDs \$61-\$80	4	Bitmask	-
\$63	Engine reference torque	2	$256*A + B$	Nm
\$66	Mass air flow sensor (dual)	5	sensor A = $(256*B+C)/32$; sensor B = $(256*D+E)/32$	g/s

Table 13.1 — Selected J1979 Mode \$01 PIDs with conversion formulas (A = first data byte, B = second, ...).

14. Manufacturer-Specific CAN Topologies

This chapter is the core of the cross-manufacturer problem. Every OEM arranges its CAN buses, module addresses, and enhanced protocols differently. Values below are the most commonly observed and widely documented; always verify against a reference vehicle. 11-bit CAN at 500 kbps is assumed unless otherwise stated.

14.1 General Motors (GMLAN)

GM's architecture is called GMLAN and consists of three physically separate CAN networks on most vehicles 2004 and later.

Bus	Bit rate	Physical	DLC Pin(s)	Typical modules
HS-GMLAN	500 kbps	Differential	6, 14	ECM, TCM, ABS/EBCM, SDM (airbag), EPS, IPC (gateway)
MS-GMLAN	95.2 kbps	Differential	3, 11 (OEM pins)	Body control, climate, radio, door, seat, memory
SW-GMLAN (LS)	33.33 kbps	Single-wire	1	Low-speed body: HVAC, seats, mirror, some lighting. Wakes bus by pulling +12 V on pin 1. High-speed mode 83.3 kbps used for programming.

Table 14.1.1 — GM GMLAN buses.

Addressing. GM uses standard 11-bit OBD-II addressing (\$7DF, \$7E0-\$7E7 / \$7E8-\$7EF) on HS-GMLAN for emissions modules. Body and chassis modules on HS-GMLAN use an OEM range where the response ID is the request ID + \$400 (e.g., request \$241 → response \$641 for BCM on some trucks). On MS-GMLAN, all modules use an OEM ID allocation, typically 11-bit IDs in the \$2xx-\$4xx range. SW-GMLAN uses a mix of IDs depending on bus master.

Enhanced protocol. GM uses Mode \$22 (Read Data by PID — note this is not UDS \$22; GM extended \$22 with manufacturer semantics on J1850 VPW and continued the name on CAN) and a UDS-compatible service set on modern Global A architecture vehicles. DTCs returned to enhanced requests can be 2-byte J1979 or 2-byte GM-specific codes — the scan tool must know which module uses which.

Eras. 1996-2005 Class 2 (J1850 VPW, 10.4 kbps) on pin 2. 2003-2007 transition vehicles have both Class 2 and HS-GMLAN. 2008+ is CAN-only for OBD-II. 2014+ Global A uses UDS, a security gateway on some premium models, and CAN-FD starting with 2019+ E2 architecture vehicles (Cadillac, Silverado/Sierra 2019+).

14.2 Ford Motor Company

Bus	Bit rate	Physical	DLC Pin(s)	Typical modules
HS-CAN	500 kbps	Differential	6, 14	PCM, TCM, ABS, RCM (airbag), steering, IPC (gateway on most)
MS-CAN	125 kbps	Differential	3, 11	BCM, SCCM (steering column), GWM, DDM/PDM (doors), RFA (keyless), APIM (SYNC), FCIM, HVAC, ACM (audio), OCSM (occupant)
FD-CAN / HS-CAN2-FD	500k / 2M	Differential	gateway internal	2018+ Lincoln, 2019+ F-series, 2020+ ADAS modules

Table 14.2.1 — Ford CAN buses.

Addressing. Ford uses standard 11-bit OBD-II addressing on HS-CAN. MS-CAN modules use OEM-specific IDs in the \$720-\$750 range for requests and \$728-\$758 for responses, but the exact pairs are per-module and per-model-year; Ford's 'network topology' table must be consulted. BCM is commonly \$726 req / \$72E resp; APIM (SYNC) is commonly \$7D0 req / \$7D8 resp on HS-CAN of newer vehicles.

Enhanced protocol. Pre-2008: Ford used proprietary Mode \$22 over J1850 PWM or ISO 14230 depending on market. 2008+: UDS on CAN. Ford's diagnostic documentation names services similarly to generic UDS, but DID assignments are Ford-specific in the \$DExx and \$4xxx ranges.

Module wake. MS-CAN is a sleep-capable bus; modules gate their own power and require a wake-up pattern (a specific CAN ID transmitted with dominant-recessive sequence) to come up. A scan tool probing MS-CAN while modules are asleep will see nothing. The convention is to ignition-on the vehicle before probing MS-CAN.

14.3 FCA / Stellantis (Chrysler / Dodge / Jeep / Ram / Fiat / Alfa)

Bus	Bit rate	Physical	DLC Pin(s)	Typical modules
CAN-C	500 kbps	Differential	6, 14	PCM/ECM, TCM, ABS/ESC, ORC (airbag), steering, TIPM/IPM
CAN-IHS (Interior High-Speed)	125 kbps	Differential	3, 11 on some	Body, radio, HVAC, instrument cluster, doors
CAN-B (body sleep)	83.3 kbps	1-wire or diff	internal	Lower-power body functions on older SRT platforms

Table 14.3.1 — FCA CAN buses.

Addressing. Standard 11-bit OBD-II on CAN-C for emissions. CAN-IHS modules use 11-bit IDs in ranges allocated by FCA per module family; the gateway (CGW / SGW) routes requests to the target bus. PCM / ECM commonly \$7E0/\$7E8; TCM \$7E1/\$7E9; ABS \$760/\$768; ORC \$751/\$759; TIPM \$740/\$748; radio \$764/\$76C.

Enhanced protocol. Pre-CAN (SCI, '1998-2004) used a proprietary 7.8 kbps bidirectional serial on pin 7 with OEM header bytes. 2005+ CAN uses mostly UDS semantics with some KWP2000-style additions. Chrysler/FCA historically used Mode \$18 for DTCs on SCI, which the scan tool still has to recognize for older diagnosis.

Security Gateway (SGW). Starting with model year 2018 on many platforms, FCA inserted a cryptographic gateway (module SGW, part of the CGW or separate) that blocks writes, actuation, routines, and even some reads until the tool authenticates. See Chapter 15 for unlock details.

14.4 Toyota / Lexus / Scion

Bus	Bit rate	Physical	Typical modules
CAN-H (chassis high-speed)	500 kbps	Differential	Engine/ECM, transmission, ABS/VSC, SRS, EPS
CAN-B (body)	500 kbps (some 250)	Differential	Body control, HVAC, meter, navigation
CAN-V (V-bus)	500 kbps	Differential	ADAS modules on newer TNGA platform
LIN sub-buses	19.2 kbps	Single-wire	Rain sensor, steering-wheel switches, door latches

Table 14.4.1 — Toyota CAN topology. Central Gateway module (CGW) routes between buses from 2016+ TNGA.

Addressing. Toyota uses standard 11-bit OBD-II on CAN-H. Body modules on CAN-B use 11-bit IDs often in the \$7C0-\$7DF (request) and \$7C8-\$7DF (response) extended range, plus \$7E0-\$7EF standard range. Toyota's DOS / Techstream tool publishes the full per-model topology.

Enhanced protocol. Toyota's implementation is largely UDS with some Toyota-specific routines under service \$31 (Routine Control) — e.g., the HV battery individual-cell check. Toyota Safety Sense / Lexus Safety System+ ADAS calibration uses service \$31 extensively.

14.5 Honda / Acura

Honda's OBD-II pre-2008 used ISO 9141-2 on pin 7, with a Honda-specific header format. From 2008 onward Honda moved to CAN. On CAN, Honda uses 11-bit OBD-II addressing; enhanced diagnostics use the Honda Diagnostic System (HDS) protocol, which layers on UDS semantics but retains some Honda-specific subservices.

Bus	Bit rate	DLC Pin(s)	Modules
F-CAN (F-bus / powertrain)	500 kbps	6, 14	ECM, TCM, VSA/ABS, EPS, SRS
B-CAN (body)	125 kbps (some 83.3)	3, 11 (OEM)	MICU (multiplex integrated control unit), gauge, HVAC, radio

Table 14.5.1 — Honda buses. Many 2008-2014 models expose only F-CAN on the DLC; B-CAN access requires aftermarket adapter or OE tool.

14.6 Nissan / Infiniti

Nissan's factory diagnostic protocol is Consult (Consult-I pre-OBD, Consult-II on ISO 9141 K-line, Consult-III and Consult-III+ on CAN). The physical layer changed from a proprietary 14-pin connector (Consult-II) to J1962 over time; modern Nissans are pure J1962 with CAN.

Bus	Bit rate	DLC Pin(s)	Modules
CAN-H	500 kbps	6, 14	ECM, TCM, ABS/VDC, airbag, EPS, meter
CAN-B	500 kbps	OEM	Body, HVAC, infotainment, keyless

Table 14.6.1 — Nissan CAN buses (typical modern).

Addressing. Standard 11-bit OBD-II on CAN-H. Nissan uses UDS-like enhanced services on newer vehicles, with manufacturer-specific DIDs in the \$1xxx-\$9xxx range and routines for common shop procedures (throttle idle relearn, CVT reset, steering angle sensor calibration, BCM self-diag).

14.7 Hyundai / Kia / Genesis

Pre-2005 Hyundai/Kia used ISO 9141-2 with a Hyundai-specific M-OBD-II profile (14229-compatible KWP2000). 2005-2008 transition: some models KWP, some CAN. 2008+: CAN with UDS on HS-CAN. Hyundai's factory tool is GDS (Global Diagnostic System).

Bus	Bit rate	DLC Pin(s)	Typical Modules
C-CAN (Chassis)	500 kbps	6, 14	ECM, TCM, HCU/ABS, SAS, EPS, airbag
B-CAN (Body)	100 kbps (some 125)	OEM	BCM, instrument cluster (integration), HVAC, TPMS
M-CAN (Multimedia)	500 kbps	OEM	Head unit, amplifier, rear seat entertainment
D-CAN (Chassis high-speed)	500 kbps	OEM	ADAS fusion, radar, camera

Table 14.7.1 — Hyundai / Kia CAN buses.

14.8 Volkswagen Group (VW / Audi / Škoda / SEAT / Porsche)

VW Group vehicles are historically notable for using **TP 2.0** (Transport Protocol 2.0) over CAN as their ISO-TP equivalent on pre-MQB platforms — it is similar in intent to ISO 15765-2 but incompatible on the wire. A generic scan tool must implement TP 2.0 to reach enhanced diagnostics on pre-2014 VW / Audi / Skoda / SEAT. 2014+ MQB and MLB platforms use standard ISO-TP and UDS.

Bus	Bit rate	DLC Pin(s)	Typical Modules
CAN-Powertrain	500 kbps	6, 14	ECM, TCM, ABS/ESP, steering, airbag
CAN-Komfort (convenience)	100 kbps	OEM / gateway	Central conv., doors, AC, seats, lighting
CAN-Infotainment	100-500 kbps	OEM / gateway	Radio/MMI, nav, amp, telephone
CAN-Extended (extended chassis)	500 kbps	OEM / gateway	4Motion, air suspension, adaptive chassis
CAN-Diagnosis	500 kbps	6, 14 (via gateway)	Gateway-exposed virtual bus for scan tool traffic

Table 14.8.1 — VW Group CAN topology. The Gateway (address \$19 on KWP legacy, \$10 on UDS) sits between the DLC and the internal buses.

Addressing and VAG 'addresses'. VAG documentation uses a two-digit hex 'address' to identify modules (01 = Engine, 02 = Transmission, 03 = ABS, 08 = HVAC, 09 = Central Electrics, 15 = Airbag, 16 = Steering, 17 = Instrument, 19 = Gateway, 25 = Immobilizer, 36 = Driver Seat, 37 = Navigation, 46 = Comfort, 55 = Headlight range, 56 = Radio, 5F = Information, 75 = Telematics, 77 = Telephone, A5 = Front Sensors, etc.). These addresses map 1:1 to CAN IDs under UDS on MQB/MLB (e.g., address 01 = request \$7E0 / response \$7E8; address 03 = \$713/\$77D; address 19 = \$710/\$77A).

Security. VW's seed/key for SA service \$27 is level-specific, with several published reverse-engineered algorithms circulating (SA level 03 for coding, level 11 for flashing on older ECUs; more levels and certificate-based schemes on MQB Evo).

14.9 BMW / Mini / Rolls-Royce

Bus	Bit rate	Physical	Typical Modules
PT-CAN (Powertrain)	500 kbps	Differential	DME/DDE (engine), EGS (trans), DSC (ABS/stab), EPS
PT-CAN2	500 kbps	Differential	F-series: added powertrain bus for more peripherals
K-CAN / K-CAN2 / K-CAN3	100 / 500 kbps	Differential	Body: KOMBI (cluster), FRM (lights), SRS, JBBF (junction-box), CIC
D-CAN / E-CAN	500 kbps	Differential	Diagnostic bus (E60 ~ E9x); DLC on newer F/G exposes D-CAN only
FlexRay	10 Mbps	Differential	F01 / F10 / F15 chassis: active roll, dampers, steering, ICM
MOST	24-50 Mbps optical	Fiber	Audio/nav ring (older iDrive)

Table 14.9.1 — BMW network topology. BMW's modules are named with three- or four-letter German acronyms which a scan tool UI should translate (e.g., KOMBI = cluster, EKPS = fuel pump, DSC = stability, EDC = electronic damper, ICM = integrated chassis, CAS = car access).

Enhanced protocols. Pre-2006: BMW DS2 over K-line, also ISTA/D uses its legacy protocol EDIABAS D-Bus. 2006+ E-series: D-CAN with KWP2000 + BMW proprietary extensions. 2014+ F/G-series: UDS on CAN and

FlexRay plus DoIP on the Central Gateway Module (ZGW / GWS). BMW CIP / ISTA/P programming sessions require specific session transitions and security access per module.

14.10 Mercedes-Benz

Bus	Bit rate	Typical Modules
CAN-C (Class C powertrain)	500 kbps	ME (engine), EGS (trans), ESP, SBC (brake), EAS (air susp)
CAN-B (Class B body)	83.3 kbps	SAM (signal acquisition), EIS (ignition), DCM (doors), KG (keyless)
CAN-D (diagnostic)	500 kbps	DLC-facing bus exposed by Central Gateway (ZGW)
CAN-E (Entertainment)	500 kbps	Head unit, NTG, amp
Most newer MRA2: CAN-FD / Ethernet backbone	2-5 Mbps / 100 Mbps	Replaces parts of the CAN hierarchy on W223 / W206 / EQS

Table 14.10.1 — Mercedes-Benz bus overview. Factory tool is Xentry (was DAS / STAR).

Enhanced protocol. Mercedes historically used a proprietary KWP2000-based protocol with module-specific framing. 2010+ moved to UDS on CAN. MRA2 (2013+ C-Class W205 onward) uses UDS, and 2018+ MRA2 vehicles typically have a security gateway and/or DoIP.

14.11 Volvo, Subaru, Mazda

Volvo. 1999-2004 used Volvo Data Link on the K-line. 2005-2014 uses CAN with Volvo's own VIDA protocol (UDS-compatible on newer platforms). 2015+ SPA and CMA use standard UDS on CAN; 2021+ SPA2 introduces Ethernet backbone and DoIP. Two CAN buses on the DLC: HS at 500 kbps (pins 6/14) and LS at 125 kbps (pins 3/11).

Subaru. Factory protocol is SSM (Subaru Select Monitor). SSM-I was K-line / TTL serial; SSM-II used a Subaru-specific 4800-bps protocol on pin 9 of an OBD-II-ish connector; SSM-III and SSM-IV use ISO 14230 (KWP2000) over K-line until 2008, then CAN. On CAN, Subaru uses 11-bit OBD-II standard with ECM at \$7E0/\$7E8, TCM at \$7E1/\$7E9; enhanced SSM requests use Subaru-specific extensions to Modes \$21 and \$22 that a generic UDS client handles correctly except for the proprietary PID semantics.

Mazda. Pre-2006 used J1850 VPW (shared with Ford pre-MNAO split) or ISO 9141 depending on market. 2006+ is CAN with Mazda's PDS (Professional Diagnostic System) protocol built on UDS with Mazda DIDs and routines. The post-2013 SkyActiv platform is fully UDS-compliant on HS-CAN.

14.12 EV-Specific Considerations (Tesla, Rivian, Lucid, BYD)

Tesla intentionally does not implement the full suite of enhanced diagnostic services accessible via the J1962 connector to third parties; Tesla's OBD-II port (where fitted) returns generic Mode \$01-\$0A and a minimal VIN, but the rich telemetry is on a Tesla-specific CAN protocol typically accessed via the center-console diagnostic port rather than the J1962. A cross-manufacturer scan tool targeting Tesla will need a Tesla-specific protocol adapter separate from J1962.

Rivian (R1T/R1S), Lucid Air, and most Chinese EVs (BYD, Nio, Xpeng) do implement J1962 + CAN + UDS for mandatory emissions / safety diagnostics, but route power and battery telemetry through an Ethernet-based zonal architecture that resembles DoIP more than classical CAN. A scan tool aiming at these should plan for DoIP support early.

15. Security Gateways and Authentication

Starting around 2018 the industry began inserting a cryptographic gateway between the J1962 connector and the internal vehicle networks. Motivations are ECU protection against tuner attacks and regulatory compliance (UN R155 / R156 cybersecurity and software-update regulations). For a scan tool this is the largest practical barrier to full cross-manufacturer capability: generic OBD-II reads still work on most vehicles, but any write, actuation, routine, or enhanced read is filtered by the gateway until the tool authenticates.

15.1 Security Architectures in Use

OEM	Gateway name	Affected MYs	Unlock mechanism
FCA / Stellantis	SGW (Security Gateway Module)	2018+ (most NA platforms) 2020+ EU	AutoAuth online service — scan tool submits VIN + subscription credentials, receives signed unlock token, presents it to SGW
GM	Global B security	2020+ on several platforms	SPS 2.0 + online GM authenticator; certificate-based session
Ford	Secure Onboard Communication (SecOC)	2018+ select, 2020+ broad	FDRS online ticketing (Ford Diagnostic & Repair System)
VW Group	Component Protection / SFD (Schutz Fahrzeug Diagnose)	2019+ MQB Evo / MLB Evo	GeKo online identity for certificate-based unlock through 5F75 / FAZIT identity; GeKo is VAG's dealer-identity system
BMW	ISTA Online backed by AOS	2015+ G-series and some F LCI	AOS / ISAP ticketing to BMW servers; short-lived per-session token
Mercedes-Benz	Xentry Pass Thru + COD	2018+ most	Xentry Pass Thru online with OEM subscription
Honda	HDS online verification	2021+ select	HDS online subscription
Toyota	No full cryptographic gateway as of MY2024 on most; but Techstream subscription required for some routines	—	Subscription-based, no strong unlock needed to read

Table 15.1 — Security-gateway status by OEM. NHTSA Right-to-Repair positioning in the US has pushed all OEMs to provide some 'independent shop' unlock path.

15.2 The UDS Service \$27 Seed / Key Handshake

Before cryptographic gateways, per-ECU security access via UDS \$27 was the standard gate. It is still in use for ECU-local operations (e.g., VIN write, immobilizer coding) even on vehicles with a cryptographic gateway in front.

```
Request: 27 N ; N odd, request seed at level (N-1)/2
Response: 67 N S1 S2 S3 S4 ; up to 4-byte seed
Request: 27 N+1 K1 K2 K3 K4 ; computed key
Response: 67 N+1 ; success
```

The key computation is an OEM- and ECU-specific function of the seed. It is typically a small XOR / rotate / table-lookup algorithm on legacy ECUs and an AES-CMAC on modern ones. Scan tools that support cross-OEM operation bundle per-OEM key-generator libraries; open-source projects (OpenSeedKey) catalog the legacy algorithms.

15.3 Session / Security Matrix

Every UDS service has a required session and security state. A scan tool must track the current state per-ECU and re-enter it after timeouts. Common requirements:

Service / DID	Required session	Required security level
\$22 (read DID) — generic	Default	None
\$22 \$F1xx — manufacturer	Extended (\$10 03)	Usually none
\$2E (write DID)	Extended or programming	Level 1 or higher
\$2F (IO control)	Extended	Level 1 (some none)
\$31 \$01 (routine start)	Extended	Varies — some need Level 1
\$34 / \$36 / \$37 (flash)	Programming (\$10 02)	Level 5 or higher
\$14 (clear DTC)	Default or extended	Usually none

Table 15.2 — Typical session / security requirements.

16. DoIP — Diagnostics over IP (ISO 13400)

ISO 13400 defines how UDS messages are carried over Ethernet instead of CAN. Any scan tool that intends to work with 2020+ luxury vehicles and most 2022+ mainstream vehicles needs a DoIP client. DoIP uses TCP for diagnostic messages and UDP for vehicle discovery, both on port 13400.

16.1 Pins and Link Activation

The vehicle must be instructed to bring up its Ethernet PHY on the DLC. ISO 13400-4 defines an **Activation Line** on DLC pin 8 which, when pulled to ground (or a specific voltage pattern per OEM), wakes the internal Ethernet bus and enables the PHY on the DLC. The Ethernet pair itself is on pins 3/11 (TX) and 12/13 (RX) on the most common scheme, but a few OEMs differ; the adapter must be electrically versatile.

After link-up, the tool obtains an IP address via DHCP (the vehicle acts as DHCP server) or uses a static IP from the vehicle's published range. It then sends a **Vehicle Identification Request** to the broadcast address on UDP 13400 and receives a **Vehicle Announcement** containing the VIN, the logical address of the gateway, the entity type, and the state of additional routing targets.

16.2 DoIP Message Types (Payload Type field)

Payload type	Message	Use
\$0000	Generic DoIP header NACK	Framing error
\$0001	Vehicle identification request	Broadcast discovery
\$0002	Vehicle identification req w/ EID	Targeted discovery
\$0003	Vehicle identification req w/ VIN	Targeted discovery
\$0004	Vehicle announcement / response	Reply with VIN, logical addr
\$0005	Routing activation request	Request to become diag client
\$0006	Routing activation response	Accepted / denied
\$0007	Alive check request	Is connection still valid?
\$0008	Alive check response	Yes
\$4001	DoIP entity status request	Max-concurrent, current sockets
\$4002	DoIP entity status response	Entity info
\$4003	Diagnostic power mode info req	Is vehicle ignition-on?
\$4004	Diagnostic power mode info resp	Power mode
\$8001	Diagnostic message	UDS payload encapsulated
\$8002	Diagnostic message positive ACK	Message delivered
\$8003	Diagnostic message negative ACK	Message rejected

Table 16.1 — DoIP payload types (ISO 13400-2).

16.3 DoIP Session Flow

1. Physical: Ethernet link up on DLC pins 3/11/12/13 after activation.
2. DHCP / IP: obtain or assign IP.
3. UDP Vehicle Identification Req broadcast → receive VIN and logical addresses of entities.
4. TCP connect to entity (gateway) on port 13400.
5. Send Routing Activation Request (payload type \$0005) with client logical address (e.g., \$0E00) and activation type \$00 (default).
6. Receive Routing Activation Response (\$0006) with result = \$10 (success).
7. Send UDS diagnostic messages encapsulated as DoIP payload type \$8001, with source = client logical address, target = ECU logical address.
8. Periodic Alive Check (\$0007) and / or UDS \$3E Tester Present to maintain session.

17. Implementation Architecture

This chapter sketches the internal architecture of a cross-manufacturer scan tool, the state machine required for protocol negotiation, and the data-model hooks that must be present to support the OEM-specific behavior described in the earlier chapters.

17.1 Layered Model

Layer	Responsibility	Key interfaces
L1 Adapter	Talk to the physical DLC hardware (ELM327, J2534, VCI, CAN-USB, etc.)	send_can_frame(), recv_can_frame(), set_baud(), set_kline_pin()
L2 Data-link	Translate between raw frames and logical PDUs for each protocol family	kline_pdu_tx/rx, j1850_pdu_tx/rx, can_pdu_tx/rx
L3 Transport	ISO-TP segmentation/reassembly; TP 2.0 on VAG legacy	send_pdu(ecu_id, bytes), recv_pdu() -> (ecu_id, bytes)
L4 Application	J1979 modes and UDS services; NRC handling	read_pid(), read_dtc(), read_did(), write_did(), routine_start(), security_access()
L5 Semantic	Convert raw DIDs / PIDs / DTCs to engineering units and localized strings using the OEM database	describe_dtc(), scale_pid(), name_module()
L6 Vehicle Identity	VIN decode, protocol detection, OEM selection, module inventory	identify_vehicle(), select_ecu_set(), session_for_task()
L7 Workflow / UI	Scan-tool user interactions — read codes, live data, bidirectional tests, procedures	App-specific

Table 17.1 — Recommended scan tool internal layering.

17.2 Vehicle Identification State Machine

```

State 1 – DISCONNECTED. Adapter not present or no DLC power.
    on adapter_up AND pin16_voltage>9 V → goto PIN_PROBE

State 2 – PIN_PROBE. Read voltage / impedance on each DLC pin.
    branch on pin results → goto PROTO_PROBE

State 3 – PROTO_PROBE. Try each candidate protocol in priority order.
    on first successful Mode $01 PID $00 response → goto VIN_READ
    on all probes failed → EXIT with error 'no compatible protocol'

State 4 – VIN_READ. Mode $09 PID $02 (fallback: UDS $22 $F190 on $7E0).
    on VIN valid → goto OEM_SELECT
    on VIN invalid / unavailable → MANUAL_PROMPT for VIN

State 5 – OEM_SELECT. Decode VIN; look up OEM topology and capabilities.
    → goto MODULE_INVENTORY

State 6 – MODULE_INVENTORY. Probe expected ECU addresses from topology.
    record present / absent / version for each
    → goto READY

State 7 – READY. Normal diagnostic operation. Periodic tester-present.
    on loss of power or tester-present timeout → goto DISCONNECTED.

```

17.3 Required Database Tables

Table	Keyed by	Columns / Purpose
wmi_to_oem	WMI (3 chars)	OEM id, region, vehicle-type hint. Plus 12-14-position WMI for small builders.
oem_vds_decoders	OEM id + model year	Function or table that, given positions 4-8 + year, returns model / body / engine / trim.
oem_topology	OEM id + platform + model year range	List of expected modules, CAN bus assignment, 11/29-bit IDs, request/response IDs, enhanced protocol (UDS vs KWP), TP variant (ISO-TP vs TP 2.0).
pid_library	Mode + PID	Description, byte count, formula, units, min/max, monitor category. Generic table plus per-OEM override tables.
dtc_library	OEM id (or 'SAE') + DTC code	Localized description, short / long text, additional-info template. Generic SAE set + per-OEM set.
did_library	OEM id + DID	Description, byte count, encoding (ASCII / BCD / raw), writability, session required, security level required.
routine_library	OEM id + routine ID	Name, description, required preconditions, step-by-step script with user prompts.
seed_key_algos	OEM id + module + security level	Key-generator function reference (compiled-in or plug-in).
doip_routing	OEM id + model year	Logical addresses of gateways and endpoints.

Table 17.2 — Database schema recommended for a generic scan tool.

17.4 Timing Requirements Summary

Parameter	Value	Notes
P2 (ISO 14229)	50 ms default, 5000 ms extended	Client timeout waiting for response. Extended applies after NRC \$78.
S3 (ISO 14229)	5000 ms default, 5000 ms extended	Server timeout; after silence the session returns to default.
Tester Present period	< S3 / 2 typical (~2 s)	Keep session alive
N_As / N_Ar / N_Bs / N_Cr	1000 ms max each	ISO-TP. Sum of N_Bs + N_Ar must stay < P2_client.
STmin (ECU-dictated)	0-127 ms or 100-900 μ s	Scan tool must honor
5-baud init (ISO 9141)	25 $\pm$ 1 ms high / 25 ms low per bit	Then 200 ms for sync
Fast init (KWP2000)	25 ms low, 25 ms high, 2 s wake-up	Then standard KWP
CAN auto-detect	200 ms per rate / variant	Stop on first success
DoIP routing activation	$\leq$ 2 s response required	Timeout drops link

Table 17.3 — Critical timing parameters.

17.5 Common Pitfalls

- **Assuming 500 kbps CAN.** Medium-duty and some European EV charging ECUs use 250 kbps. Always probe both if no response at 500.

- **Forgetting functional broadcast responses can collide.** When multiple ECUs respond to \$7DF, each response may overlap in time. The scan tool must gather all responders with a >200 ms window, then disambiguate by source CAN ID.
- **Ignoring NRC \$78 (response pending).** This is not an error. It means the ECU needs more time; keep waiting (extended P2) until a positive or another negative arrives.
- **Not suppressing response on \$3E.** Tester Present every 2 s is chatty. Use \$3E 80 (sub-function with suppressPosResp bit) to cut bus traffic in half.
- **Leaking session.** Entering programming session but failing to return to default on disconnect can leave the vehicle in a state where the driver gets warning lights until the next key-cycle.
- **Mishandling VAG TP 2.0.** Pre-MQB VWs use TP 2.0 framing rather than ISO-TP, and require a Channel Open / Channel Setup handshake (addresses \$200 / \$201, module address as tester-selectable) before any KWP request. Scan tools that speak only ISO-TP silently fail to read these ECUs.
- **Missing MS-CAN wake-up on Ford.** MS-CAN modules sleep. Probing with ignition off yields nothing. Cycle key on before probing.
- **Security Gateway silently blocking enhanced services.** On 2018+ FCA, \$22 \$F1 \$90 still reads VIN, but \$2F (IO control) and \$31 (routine start) return \$7F \$xx \$33 (security denied) or \$7F \$xx \$31 (requestOutOfRange) until AutoAuth unlock.
- **Treating P0, P2, and P3 ranges as identical.** P0xxx is generic; P1xxx is OEM; P2xxx is generic (extended); P3xxx is OEM-specific for Asian OEMs, generic for some newer codes. Display exactly as J2012 specifies — don't strip a leading zero.
- **Checksum / CVN not displayed.** Emissions inspections rely on Mode \$09 PID \$04 (CAL ID) and PID \$06 (Calibration Verification Number). Tools that skip these fail inspection-support use cases.

18. Appendices — Lookup Tables

Appendix A — Complete DLC Pin Summary

Pin	Most common assignment	Less common uses
1	GM SW-CAN, some OEM discretionary	Ford FEPS (programming voltage, legacy)
2	J1850 Bus+	—
3	Ford MS-CAN High, DoIP Ethernet TX+	Chrysler discretionary
4	Chassis ground	—
5	Signal ground	—
6	HS-CAN High (ISO 15765-4)	—
7	K-line (ISO 9141-2, ISO 14230)	Subaru SSM-II legacy
8	DoIP Activation Line	Ford MS-CAN Low on some
9	OEM discretionary	Subaru SSM-II legacy signal
10	J1850 Bus-	—
11	Ford MS-CAN Low, DoIP Ethernet TX-	Chrysler CCD on very old
12	DoIP Ethernet RX+	OEM discretionary
13	DoIP Ethernet RX-	OEM discretionary
14	HS-CAN Low (ISO 15765-4)	—
15	L-line (ISO 9141-2 wake-up)	Rarely used
16	Battery +12 V (unswitched)	—

Appendix B — Standard Baud Rates

Protocol	Baud rate(s)
HS-CAN (cars)	500 kbps
HS-CAN (commercial)	250 kbps
MS-CAN	125 kbps
SW-CAN (GMLAN LS)	33.33 kbps normal; 83.3 kbps high-speed mode
Mercedes CAN-B	83.3 kbps
VW CAN Komfort / Infotainment	100 kbps
J1850 VPW	10.4 kbps
J1850 PWM	41.6 kbps
ISO 9141-2 / ISO 14230	10.4 kbps typical (varies 1.2-125 kbps)
CAN-FD (arbitration / data)	500 kbps / 2 Mbps typical; 500k/5M on some
FlexRay (BMW ICM, some Audi)	10 Mbps
DoIP / automotive Ethernet	100 Mbps BroadR-Reach; 1 Gbps on newest

Appendix C — Generic Mode \$01 Supported-PID Chain

Each of these PIDs, when its bit is set in a prior supported-PID response, itself returns the supported-PID bitmask for the next range.

PID	Covers	Returned if bit set in
\$00	\$01-\$20	(always request first)
\$20	\$21-\$40	\$00
\$40	\$41-\$60	\$20
\$60	\$61-\$80	\$40
\$80	\$81-\$A0	\$60
\$A0	\$A1-\$C0	\$80
\$C0	\$C1-\$E0	\$A0

Appendix D — Key Standards Reference

Standard	Title / Scope
ISO 3779	VIN — content and structure
ISO 3780	WMI — world manufacturer identifier
SAE J1962	Diagnostic connector (DLC)
SAE J1850	Class B data communications (PWM / VPW)
SAE J1979 / ISO 15031-5	E/E diagnostic test modes (generic OBD)
SAE J2012 / ISO 15031-6	DTC definitions
SAE J2178	Class B application message definitions (J1850 era)
SAE J2190	Enhanced diagnostic test modes (legacy enhanced)
SAE J2284	HS-CAN physical and data-link
SAE J2411	Single-wire CAN physical layer
SAE J2534	Pass-Thru reprogramming interface
ISO 9141-2	K-line, 10.4 kbps OBD-II
ISO 11898-1 / -2	CAN data link / physical
ISO 13400	DoIP — Diagnostics over IP
ISO 14229	UDS — Unified Diagnostic Services
ISO 14230	KWP2000
ISO 15031	Communication between vehicle and external test equipment
ISO 15765-2	ISO-TP — transport protocol for CAN
ISO 15765-4	OBD-II variants of CAN
ISO 22900 (MVCI, RP1210)	Modular vehicle communication interface
ISO 22901 (ODX)	Open Diagnostic Data Exchange format
UN ECE R155 / R156	Cybersecurity and software-update regulations

Appendix E — Suggested Implementation Priorities

A scan tool team with limited budget can reach broad vehicle coverage by implementing features in roughly this order:

Phase	Scope	Coverage achieved
1	ISO 15765-4 CAN 11-bit 500 kbps + J1979 Modes \$01-\$0A	~85 % of NA vehicles 2008+; all basic OBD-II use cases
2	VIN decode with full WMI table; per-OEM model-year tables	Reliable OEM identification
3	UDS \$10, \$22, \$19, \$27, \$31, \$3E, \$85 for enhanced reads	Enhanced reads on most 2010+ vehicles
4	OEM topology tables + module naming for top 8 OEMs	Bidirectional basic tests and procedures
5	KWP2000 / ISO 9141-2 / J1850 for legacy coverage	1996-2007 vehicles

Phase	Scope	Coverage achieved
6	VW TP 2.0, Ford MS-CAN, BMW D-CAN legacy	Enhanced access on pre-2015 EU vehicles
7	DoIP + routing activation	2020+ luxury and modern EVs
8	Security gateway unlock (FCA AutoAuth, VW SFD, Ford FDRS, GM SPS)	Writes / routines on 2018+ vehicles
9	Manufacturer seed/key libraries for module-local \$27	VIN write, immobilizer, calibrations
10	Flashing (\$34/\$36/\$37) and CAN-FD physical	Full programming capability

Table 18.1 — Recommended build-out order.

Appendix F — Closing Note

Every row in this document is a starting point. The vehicle in front of the tool is the ground truth. When coverage surprises emerge — and they will — the right answer is almost always to capture a CAN trace of the OE tool performing the same operation, diff it against your tool's behavior, and update the per-OEM tables. The public standards are necessary but never sufficient for cross-manufacturer capability; observation of real vehicles is the remaining ingredient.